Reanalysis converts an archived PL-Seq study into a reusable research resource

Reproduction, statistical refinement and read-level visualization were integrated into one executable workflow.

A

From archived publication to reusable analysis

Published study	Recover	Recompute	Reanalyze	Reuse
Paper, supplements, SRA accessions and BEDs	Map samples; audit files; reconstruct missing context	FASTQ -> dm6 BAM; long/short molecule filters	Replicate-level effects; BH-adjusted q values	Read-level figures, Quarto, container and data products

Human review at every stage: biological questions, validation, interpretation and final presentation

B

What the reanalysis adds

	Published representation	Reusable reanalysis product
Evidence	Summary PSI	Replicate-resolved PSI + observed molecules
Statistics	Nominal significance / stars	Effect sizes + BH-adjusted q values
Interpretation	Pooled changes	Genotype-by-3'UTR interaction
Presentation	Manually assembled figures	Code-generated main and locus-catalog figures
Reproducibility	Static publication outputs	Apptainer, Slurm, manifests, checksums and Quarto

C

A three-day intensive reanalysis sprint

24-26 October 2026

Day 1	Day 2	Day 3
Paper + supplement audit Phase 1/2 tables FASTQ reconstruction pilot	HPC alignment + filtering Replicate-aware statistics Figures 2-4 redesign	Figure 5 + locus catalog Publication-style figure system Manuscript architecture